



# Eventard

Campus event board with live RSVPs and Add to Google Calendar

A 300-level project for the University of Uyo. Students browse events, RSVP and save them to their calendar. Student Affairs publishes events that appear on every open board within seconds.

**38**

seeded events

**50**

users (2 admins)

**1092**

RSVPs

## Contents

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Demo logins: student@uniuyo.edu.ng / student123 and admin@uniuyo.edu.ng / admin123. Screens captured with the clock frozen at 2026-10-05.

# 1. Overview

Students miss campus events because announcements are scattered across WhatsApp groups and notice boards. Eventard puts them on one board.

| Who                     | What they can do                                                                                                                                                           |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Student                 | Browse upcoming and past events, search and filter by category, RSVP or cancel, see the live count and who is going, add an event to Google Calendar, see their own RSVPs. |
| Admin (Student Affairs) | See dashboard numbers, publish events with a banner (upload or library), edit, cancel, restore or delete events, and see the full attendee list.                           |

## Features

- Live RSVP count without a page reload (server re-render every 4 to 6 seconds).
- Capacity limits that cannot be overbooked, even when students click at the same moment.
- Add to Google Calendar with no API key.
- Full, almost full, live, ended and cancelled states.
- Inline form validation with Zod.
- Works on phones.

## Design

The interface follows the visual style of Claude: warm ivory background, serif headings (Source Serif 4), a clean sans body (Instrument Sans), soft borders and a terracotta accent. It is a normal web app, not a chatbot.

## 2. How the core logic works

All business rules are pure functions in `src/lib/`. They take "now" as an argument, so the tests and the demo can freeze the clock with `EVENTARD_TODAY=YYYY-MM-DD` (the clock is then 10:00 Lagos time on that day).

### Event phase

```
eventPhase(event, now)
  cancelled  if status is cancelled
  upcoming   if now < starts_at
  live       if starts_at <= now < ends_at
  ended      otherwise
```

### Can this student RSVP?

```
rsvpState(event, goingCount, hasRsvp, now)
  cancelled or ended  -> closed
  hasRsvp             -> "going" (button cancels)
  live                -> closed for new RSVPs
  capacity reached    -> "full"
  otherwise           -> "open"
```

### No overbooking

`toggleRsvp` runs in a database transaction. It first runs `SELECT ... FOR UPDATE` on the event row, so two requests for the same event wait for each other. It then counts RSVPs, applies `rsvpState` and inserts or deletes. A unique constraint on `(event_id, user_id)` stops duplicates. The integration test fires 10 RSVPs at a 3 seat event at once and checks exactly 3 succeed.

### Add to Google Calendar

```
https://calendar.google.com/calendar/render?action=TEMPLATE
&text=Code%20and%20Coffee%3A%20Build%20a%20Web%20App%20with%20Next.js
&dates=20261006T130000Z/20261006T160000Z
&details=...&location=ICT%20Centre%2C%20Lab%202%2C%20Town%20Campus
```

Times are stored as `timestamp tz`. `toGoogleDate` converts to UTC and strips punctuation, so 2:00 pm Lagos becomes 130000Z. Every text field goes through `encodeURIComponent`.

### Live updates

A tiny client component, `LiveRefresh`, calls `router.refresh()` on a timer while the tab is visible. Next.js re-renders the Server Components and sends only the changed parts, so the count and new events appear without a reload. Server Actions also call `revalidatePath` so the person who clicked sees the change at once.

### Sessions

Passwords are hashed with `bcryptjs`. On sign in the server sets a JWT signed with HS256 (jose) in an HTTP-only, `SameSite=Lax` cookie. `src/proxy.ts` redirects visitors without a valid token, and every page and Server

Action checks the session again on the server.



### 3. Architecture and data model

#### Browser

Server-rendered pages, small client components for forms, RSVP and live refresh.

#### Next.js16 (App Router)

Server Components read data. Server Actions change it. proxy.ts guards routes. /api/health pings the database.

#### PostgreSQL

Drizzle ORM with versioned SQL migrations from drizzle-kit.

| Folder          | Contents                                                                                                                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| src/app         | Routes: /login, / (board), /events/[id], /my, /admin, /admin/events/new, /admin/events/[id]/edit, /api/health, /api/images/[id] |
| src/app/actions | Server Actions: login, logout, RSVP, create, update, cancel, delete                                                             |
| src/lib         | Pure rules (events.ts, calendar.ts, today.ts, format.ts), validation, auth, queries                                             |
| src/db          | Drizzle schema, connection pool, seed data                                                                                      |
| drizzle/        | Generated SQL migrations                                                                                                        |
| tests/, e2e/    | Vitest unit and integration tests, Playwright end-to-end tests                                                                  |

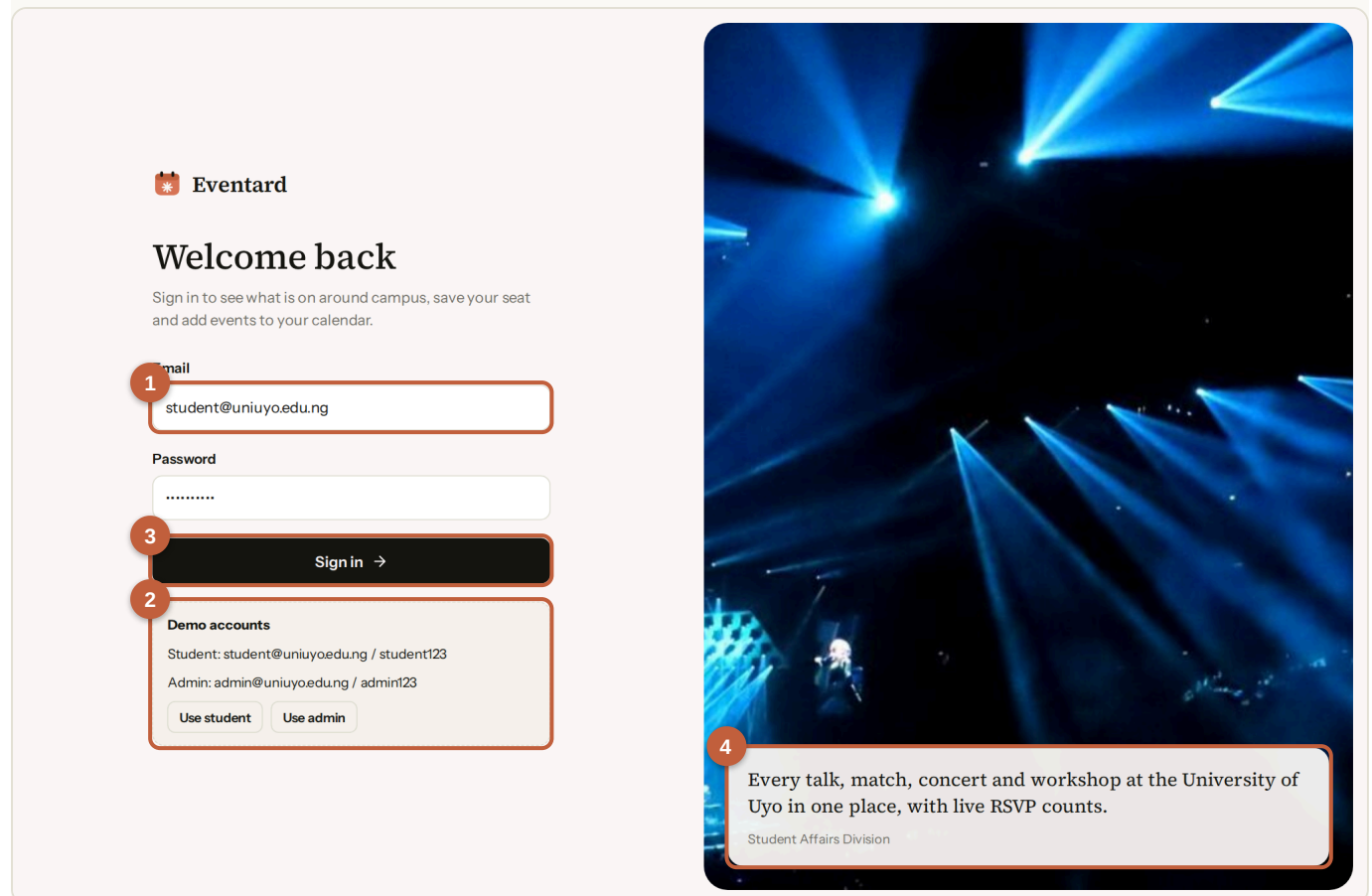
#### Tables

| Table   | Columns                                                                                                           | Rules                                                                       |
|---------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| users   | id, name, email, password_hash, role (student or admin), department, level, created_at                            | email unique                                                                |
| events  | id, title, description, category, venue, starts_at, ends_at, banner_url, capacity, status, created_by, created_at | ends_at > starts_at, capacity null or positive, created_by references users |
| rsvps   | id, event_id, user_id, created_at                                                                                 | unique (event_id, user_id), cascades on delete                              |
| uploads | id (uuid), mime, data (bytea), uploaded_by, created_at                                                            | Banners uploaded by admins, served by /api/images/[id]                      |

Uploaded banners are kept in Postgres rather than on disk because Railway containers have no persistent file system.

## 4. Screen walkthrough

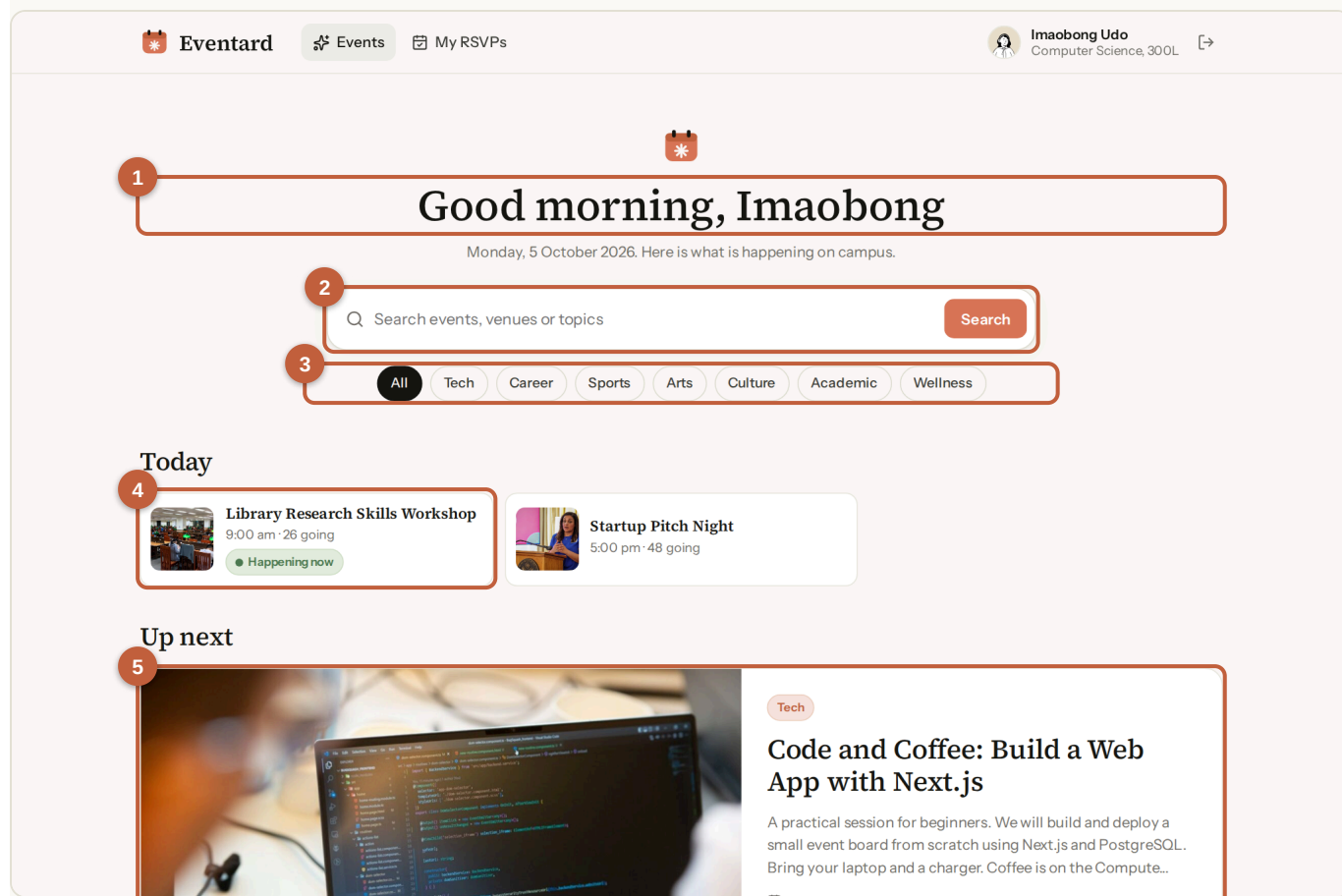
### Sign in



*Sign-in page with the demo login prefilled.*

- 1 Email field, prefilled with the demo student account so the presenter can sign in with one click.
- 2 Demo accounts panel. The Use admin button swaps the fields to the Student Affairs admin login.
- 3 Sign in posts to a Server Action that checks the bcrypt hash and sets a signed JWT in an HTTP-only cookie.
- 4 Brand panel with a real campus concert photo (Creative Commons, stored locally).

## Event board



*The board as the demo student sees it.*

- 1 Personal greeting using the student's first name and the campus time of day (Africa/Lagos).
- 2 Search box. Matches event titles, venues and descriptions with a case-insensitive query.
- 3 Category filter chips. The active chip is dark; the filter is kept in the URL so it can be shared.
- 4 Today strip: events on the current campus day with their start time and live RSVP count.
- 5 Up next: the nearest upcoming event, shown large so the demo starts from it.

## Upcoming events grid

Coming up

1

Upcoming

Past

2

OCT 7



Career

✓ Going

**Career Fair 2026: Meet 30 Employers**

🕒 Wed, 7 Oct, 9:00 am

📍 Main Auditorium, Main Campus

4



46 going

3

OCT 8



Sports

**Inter-Faculty Basketball Semi Final**

🕒 Thu, 8 Oct, 4:00 pm

📍 Indoor Sports Hall

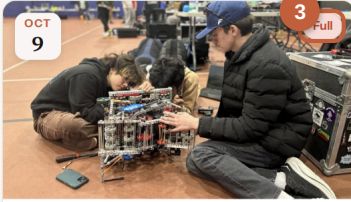


33 going

of 200

3

OCT 9



Tech

**Robotics Club Open Day**

🕒 Fri, 9 Oct, 11:00 am

📍 Faculty of Engineering Workshop



45 going

of 45

OCT 10



Wellness

**Mental Health Awareness Talk**

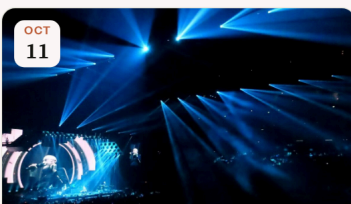
🕒 Sat, 10 Oct, 1:00 pm

📍 Faculty of Social Sciences Hall



19 going

OCT 11



Arts

✓ Going

**Afrobeats Live: Campus Concert**

🕒 Sun, 11 Oct, 6:00 pm

📍 Town Campus Field



47 going

OCT 12



Career

**Design Thinking Workshop**

🕒 Mon, 12 Oct, 10:00 am

📍 Entrepreneurship Centre, Room 3



31 going

of 35

Event cards with status flags and attendee avatars.

- 1 Upcoming and Past tabs. Past events keep their RSVP history but cannot be booked.
- 2 Date badge in campus time.
- 3 Status flag computed by eventPhase and isAlmostFull: Full, a few spots left, Happening now or Cancelled.
- 4 Avatar stack of the latest attendees with the total going. Avatars are generated locally with DiceBear.

## Event page

Eventard

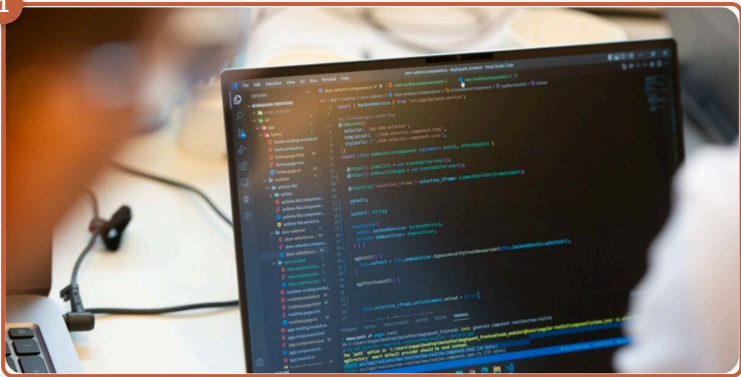
Events

My RSVPs

Imaobong Udo  
Computer Science, 300L

All events

1



Tech Tomorrow

# Code and Coffee: Build a Web App with Next.js

2

Tuesday, 6 October 2026

Date

2:00 pm to 5:00 pm

West Africa Time

ICT Centre, Lab 2, Town Campus

Venue

60 seats

Capacity

A practical session for beginners. We will build and deploy a small event board from scratch

3

Live count

60 seats

37

students going

4

+ RSVP to this event

5

Add to Google Calendar

Who is going

Halima Abubakar

Ruth Ekong

Daniel Ebong

Ngozi Obi

and 33 others

Event page before the student RSVPs.

- 1 Event banner.
- 2 Date, time range (West Africa Time), venue and capacity.
- 3 Live count. The page calls `router.refresh()` every 4 seconds so other students' RSVPs show up without a reload.
- 4 RSVP button. It runs a Server Action that locks the event row, re-checks capacity and inserts the RSVP.
- 5 Add to Google Calendar. A plain link with the title, UTC dates, description and venue URL-encoded. No Google API key is needed.

## After RSVP

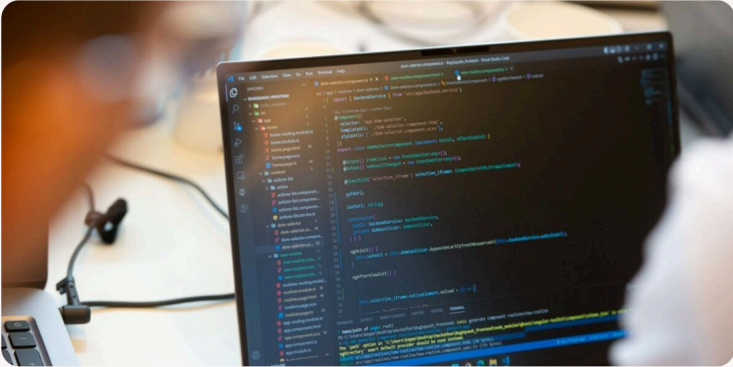
Eventard

Events

My RSVPs

Imaobong Udo  
Computer Science, 300L

← All events



Tech Tomorrow

### Code and Coffee: Build a Web App with Next.js

Tuesday, 6 October 2026  
Date

2:00 pm to 5:00 pm  
West Africa Time

ICT Centre, Lab 2, Town Campus  
Venue

60 seats  
Capacity

A practical session for beginners. We will build and deploy a small event board from scratch

Live count 60 seats

1 38 students going

3

2

✓ You are going. Tap to cancel!

Add to Google Calendar

Who is going

Halima Abubakar

Ruth Ekong

Daniel Ebong

Imaobong Udo

and 34 others

The same page right after clicking RSVP.

- 1 The count went from 37 to 38 straight after the click.
- 2 The button turns green. Tapping it again cancels the RSVP and frees the seat.
- 3 Capacity bar: 38 of 60 seats taken.



# My RSVPs

 **Eventard**

Events

My RSVPs

 **Imaobong Udo**  
Computer Science, 300L

1

## My RSVPs

5 upcoming, 4 attended. Only you can see this list.

Find more events

### Upcoming

OCT 5



Career

Going

#### Startup Pitch Night

Mon, 5 Oct, 5:00 pm

Entrepreneurship Centre Hall

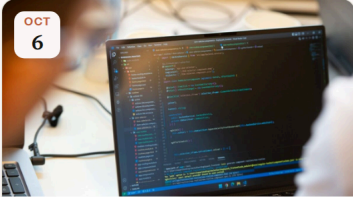
48 going

of 120

2

Add to Google Calendar

OCT 6



Tech

Going

#### Code and Coffee: Build a Web App with Next.js

Tue, 6 Oct, 2:00 pm

ICT Centre, Lab 2, Town Campus

38 going

of 60

Add to Google Calendar

OCT 7



Career

Going

#### Career Fair 2026: Meet 30 Employers

Wed, 7 Oct, 9:00 am

Main Auditorium, Main Campus

46 going

Add to Google Calendar

OCT 11



OCT 16



The student's own bookings.

- 1 My RSVPs only ever queries rows where user\_id is the signed-in student.
- 2 A calendar button under each upcoming booking.
- 3 Going badge.

## Admin dashboard

 **Eventard**

Events

Dashboard

 **Dr. Uwem Ekpennyong**  
Student Affairs

Events dashboard

Publish events and watch RSVPs come in.

1

23

Upcoming events

9

In the next 7 days

603

RSVPs for upcoming events

1

Upcoming events at capacity

2

+ New event

Upcoming events

5

Upcoming

Past

| EVENT                                                                                                                                                                    | WHEN                   | RSVPS                           | STATUS |                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------|--------|---------------------------------------------|
|  <b>Library Research Skills Workshop</b><br>University Library Seminar Room             | Mon, 5 Oct<br>9:00 am  | <div>26 / 30</div> <div>3</div> | Live   | <button>View</button> <button>Edit</button> |
|  <b>Startup Pitch Night</b><br>Entrepreneurship Centre Hall                             | Mon, 5 Oct<br>5:00 pm  | <div>48 / 120</div>             | Open   | <button>View</button> <button>Edit</button> |
|  <b>Code and Coffee: Build a Web App with Next.js</b><br>ICT Centre, Lab 2, Town Campus | Tue, 6 Oct<br>2:00 pm  | <div>38 / 60</div>              | Open   | <button>View</button> <button>Edit</button> |
|  <b>Career Fair 2026: Meet 30 Employers</b><br>Main Auditorium, Main Campus             | Wed, 7 Oct<br>9:00 am  | <div>46</div>                   | Open   | <button>View</button> <button>Edit</button> |
|  <b>Inter-Faculty Basketball Semi Final</b><br>Indoor Sports Hall                       | Thu, 8 Oct<br>4:00 pm  | <div>33 / 200</div>             | Open   | <button>View</button> <button>Edit</button> |
|  <b>Robotics Club Open Day</b><br>Faculty of Engineering Workshop                      | Fri, 9 Oct<br>11:00 am | <div>45 / 45</div> <div>4</div> | Full   | <button>View</button> <button>Edit</button> |
|  <b>Mental Health Awareness Talk</b><br>Faculty of Social Sciences Hall               | Sat, 10 Oct<br>1:00 pm | <div>19</div>                   | Open   | <button>View</button> <button>Edit</button> |

Student Affairs dashboard.

- 1 Dashboard numbers: upcoming events, events in the next 7 days, RSVPs for upcoming events and events at capacity.
- 2 New event button.
- 3 Fill bar for events with a capacity. It turns red at 100 percent.
- 4 Full and Cancelled states are easy to spot.
- 5 Switch between upcoming and past events.



## Publish an event

Eventard

Events

Dashboard

Dr. Uwem Ekpennyong  
Student Affairs

← Dashboard

### Publish an event

It appears on every student's board as soon as you save.

1

Please fix the highlighted fields.

2

Title  
e.g. Career Fair 2026  
Title must be at least 4 characters

3

Category  
Choose a category  
Pick a category

Venue  
e.g. Main Auditorium, Main Campus  
Enter a venue

Starts  
10/08/2026, 02:00 PM

Ends  
10/08/2026, 04:00 PM

Capacity  
Leave empty for no limit

Description  
What will happen, who it is for and what to bring  
Describe the event in at least 20 characters

4

Banner  
Upload a photo (JPG, PNG or WebP, up to 3 MB) or pick one from the library.

Upload image

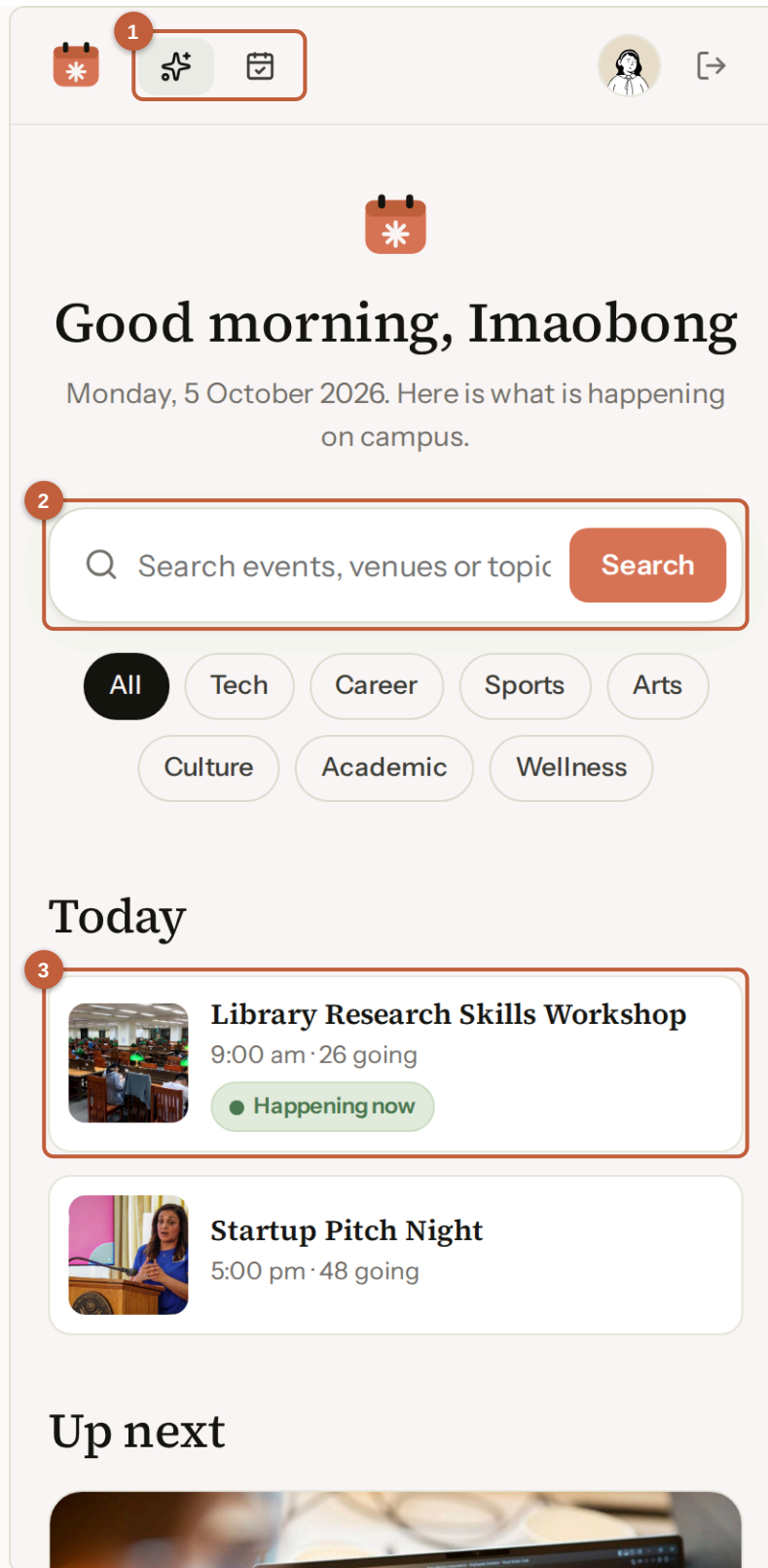


New event form after submitting it empty.

- 1 Summary message when the form has problems.
- 2 Inline field errors from the Zod schema, shown under each input.
- 3 Start and end use datetime-local inputs in campus time. Business rules reject past starts, ends before starts and events longer than 3 days.
- 4 Banner library of real photos. Admins can also upload a JPG, PNG or WebP up to 3 MB, which is stored in Postgres.

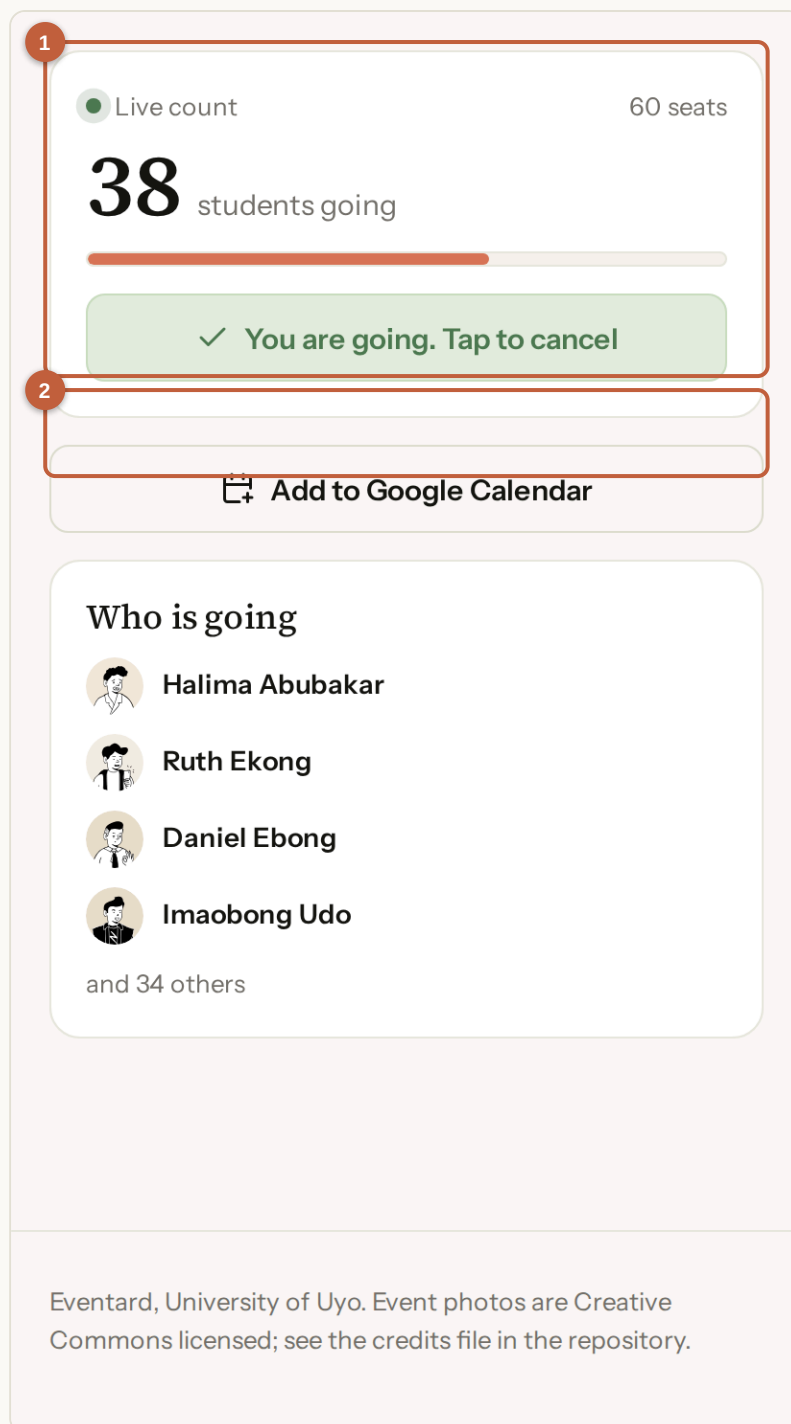
## 5. Mobile view

Layouts use CSS grid with `minmax(0, 1fr)` columns so long titles and tables never push the page wider than the screen. The mobile e2e test runs on a Pixel 7 profile and fails if the page scrolls sideways.



*Board on a phone.*

- 1 On phones the navigation collapses to icons.
- 2 Search stays full width.
- 3 Cards stack into one column; no sideways scrolling (checked in the mobile e2e test).



*RSVP panel and calendar button on a phone.*

- 1 On mobile the RSVP panel follows the event details.
- 2 Calendar button, full width for easy tapping.

## 6. Running locally

```
service postgresql start
sudo -u postgres createdb eventard
sudo -u postgres createdb eventard_test
cp .env.example .env          # set DATABASE_URL and SESSION_SECRET
npm install
npm run db:migrate
npm run db:seed                # resets and seeds demo data relative to today
npm run dev                    # http://localhost:3000
```

| Script                                          | What it does                                                                            |
|-------------------------------------------------|-----------------------------------------------------------------------------------------|
| npm run dev / build / start                     | Next.js development server, production build, production server                         |
| npm run start:prod                              | Migrate, seed only if the database is empty, then next start on 0.0.0 (used by Railway) |
| npm run db:generate                             | Create a new SQL migration after editing src/db/schema.ts                               |
| npm run db:migrate / db:seed                    | Apply migrations / reset and seed                                                       |
| npm test, test:unit, test:integration, test:e2e | Run the test suites                                                                     |
| npm run docs:pdf                                | Rebuild this PDF                                                                        |

Set `EVENTARD_TODAY=2026-10-05` to freeze the clock for a rehearsal.

# 7. Testing

42

unit tests passed

12

integration tests passed

7

e2e tests passed

| Suite       | Tool                                    | What it covers                                                                                                                                                                                                                      |
|-------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit        | Vitest                                  | Event phases, RSVP rules, capacity helpers, campus day labels across UTC midnight, time validation, Google Calendar URL format and encoding, datetime-local conversion, Zod schemas, image checks, JWT signing and tampering.       |
| Integration | Vitest + real Postgres (eventard_test)  | Migrations and seed, RSVP toggle, full/past/cancelled refusals, unique constraint, 10 concurrent RSVPs for 3 seats, per-user scoping, search and category filters, admin stats and attendees, the end-after-start check constraint. |
| End to end  | Playwright (desktop Chrome and Pixel 7) | Login redirect and prefill, wrong password, RSVP and live count, calendar link parameters, full event, inline form errors, admin publishes and a student's open board picks it up, mobile RSVP with no sideways scroll.             |

Latest run: unit 42, integration 12, 0 failed; e2e 7 of 7 passed (6 desktop, 1 mobile).

```
npm run test:unit
npm run test:integration # needs eventard_test
npm run test:e2e         # builds the app, seeds eventard_test with EVENTARD_TODAY=2026-10-05
```

## 8. Deployment

Eventard runs on Railway in the **school-projects** project as its own service with its own PostgreSQL database (**eventard-postgres**). The GitHub repository is connected to the service, so every push to main deploys.

| Setting        | Value                                                              |
|----------------|--------------------------------------------------------------------|
| Build          | npm run build                                                      |
| Start          | npm run start:prod (migrate, seed if empty, next start -H 0.0.0.0) |
| Health check   | /api/health (returns 503 if the database cannot be reached)        |
| DATABASE_URL   | \${{eventard-postgres.DATABASE_URL}}                               |
| SESSION_SECRET | random 48 byte value                                               |
| NODE_ENV       | production                                                         |

These settings live in `railway.json` in the repository.

## 9. Five minute presentation script

| Time | What to do and say                                                                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 | Problem: event news is scattered across WhatsApp groups and notice boards, so students miss talks and fairs. Eventard is one board with RSVPs and calendar reminders.                             |
| 0:30 | Open the sign-in page. Point out the prefilled demo login. Sign in as the student. Show the greeting, the Today strip and the Up next card.                                                       |
| 1:15 | Search for "hackathon", then click the Tech chip. Clear the filter. Point out Full and spots-left flags on cards.                                                                                 |
| 1:45 | Open "Code and Coffee". Say the count is 37. Click RSVP: it jumps to 38 and the button turns green. Explain the row lock that stops overbooking.                                                  |
| 2:30 | Click Add to Google Calendar. Google opens with title, time, venue and description filled in. Explain the link is built from the event data, dates in UTC, no API key.                            |
| 3:00 | Open My RSVPs to show only this student's bookings.                                                                                                                                               |
| 3:20 | In a second window sign in as admin. Show the dashboard numbers and the red Full bar for Robotics Club Open Day.                                                                                  |
| 3:45 | Click New event, submit empty to show inline errors, then fill it in, pick a banner and publish. Switch to the student window: the new event appears within a few seconds without reloading.      |
| 4:30 | Architecture in one sentence each: Next.js Server Components and Server Actions, Drizzle with PostgreSQL, JWT cookie sessions, tests (quote the counts), deployed on Railway with a health check. |
| 4:50 | Close: what we would add next (email reminders, QR check-in at the door). Take questions.                                                                                                         |



## Image credits

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